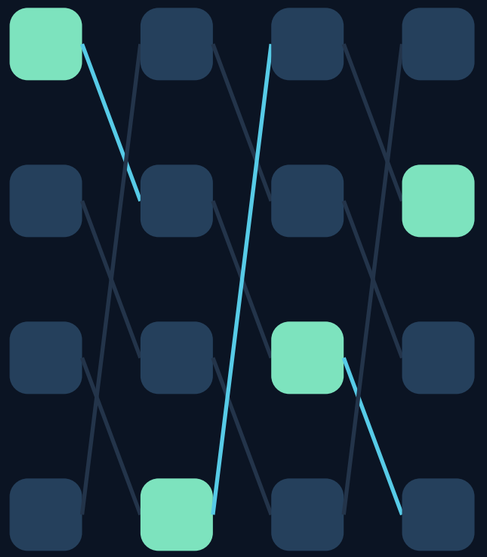
# Sleeping Machines

The next general-purpose substrate for machine intelligence.

One learner. Many forms of experience. Reusable intelligence, from representations to execution.

**Tero Keski-Valkama | Sole founder**

Raising €3M to scale research and engineering
Research-stage deep tech · 4 October 2026



**Capacity beyond activity**

# One substrate for learning, representing and executing intelligence.

Different forms of experience can teach reusable computations within the same learner.

**MANY FORMS OF EXPERIENCE**

## Observe + act

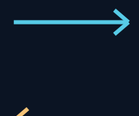
Tokens, sensors, typed tables and process traces. Synchronous samples and asynchronous events.



**ONE TRAINABLE SUBSTRATE**

## Learn + remember

Addressed messages, persistent memory and temporal races. End-to-end credit learns content, timing and unrealized alternatives.



**REUSABLE INTELLIGENCE**

## Compose + transfer

Transfer between perception, reasoning and action. Continual learning and self-design.

**The platform thesis: knowledge learned from one form of experience can become useful computation in another.**

Time and sleeps perform computation. Target: clockless learning and inference, capacity beyond activity, small data to large scale.

**First bridge: mixed-type tables through the temporal core. First transfer proof: joint learning, retained skills, matched work.**

# Nine public leaderboard wins, and language is next.

Official splits, sealed test sets scored once per run, the published state of the art as the bar. 40 of 41 independent runs ahead.

### EVENT PREDICTION

## 5 of 5

All five EasyTPP datasets won, and one configuration with no per-dataset tuning wins all five (24 of 25 seeds ahead).

### FAR SMALLER

## up to 15×

Smaller and cheaper than the state of the art (Retweet 1/15, Taxi 1/12, Amazon 0.29×), while more accurate.

### NEW DOMAINS

## 4 more

P19 sepsis (AUPRC 0.639 vs 0.583), PAM activity (0.978 vs 0.975), TGB trade (0.868 vs 0.863) and TGB wiki (0.835 vs 0.827).

### LANGUAGE

## lead grows

Tokenized text: ahead of a Kneser–Ney trigram (6.009 vs 6.537 nats/token), and further ahead at 4× the data.

**One configuration wins all five EasyTPP datasets; both temporal-graph benchmarks won at the first sealed attempt.**

# Public-leaderboard wins: timed event data (EasyTPP).

Public benchmark (ICLR 2024), official splits. Log-likelihood per event in nats: higher is better. Sealed test, 5 seeds.

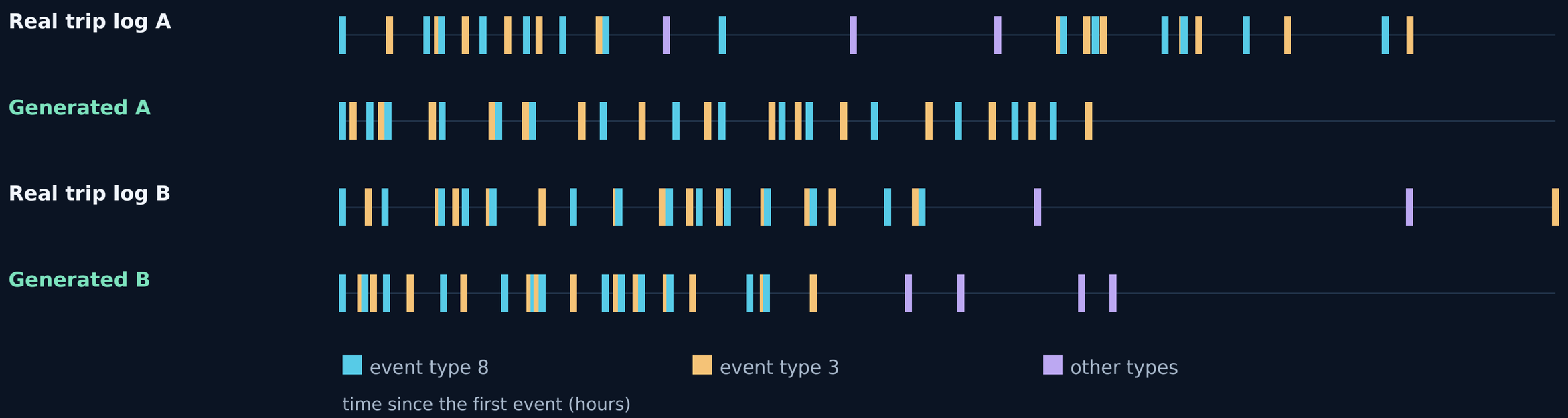
| Dataset | Best published | Ours | Result and compute we used |
|---|---|---|---|
| Taobao (shopping) | 1.318 | 1.399 ± 0.003 | Win · +0.081 nats/event at 0.92× the leader’s compute |
| Taxi (mobility), 5-seed mixture | 0.522 | 0.536 | Win · beats every published model at 0.41× its compute |
| Taxi, single model | 0.522 ± 0.004 | 0.525 ± 0.001 | Ahead on the mean at 1/12 of its parameters and compute |
| StackOverflow (Q&A activity) | −2.163 ± 0.009 | −2.144 ± 0.004 | Win · +0.019 nats/event at 1.26× the leader’s compute |
| StackOverflow, matched size | −2.163 ± 0.009 | −2.153 ± 0.005 | Win · all 5 seeds ahead at 1.0× the leader’s compute |
| Amazon (shopping reviews) | 0.781 ± 0.011 | 0.803 ± 0.001 | Win · all 5 seeds ahead at 0.29× the leader’s compute |
| Retweet (social) | −6.348 | −6.326 ± 0.001 | Win at 1/15 of S2P2's compute; all 5 seeds ahead |

**All five datasets won, each at about the state of the art’s compute or far less.**

Leaders: S2P2 on Taxi, StackOverflow, Amazon; IFTPP on Taobao; NHP on Retweet. Compute = per-event multiply-adds. Taxi reproduced on separate hardware.

# The same model also generates realistic event streams.

Taxi trips: real held-out sequences and sequences generated by racing the model's clocks from the same first event.




Over 200 held-out sequences: gap distribution KS distance 0.014 (naive reference 0.063); event-to-event transitions 0.024 total variation (naive 0.53).

**Exact sampling, no rejection: simulation, what-if analysis and synthetic data from the winning model.**

Naive reference: event types from overall frequencies, exponential gaps with the real mean. Single model, seed 0; examples drawn with a fixed seed.

# Public-benchmark win: ICU sepsis prediction (P19).

PhysioNet 2019, 38,803 ICU stays, 34 irregularly sampled channels, 4.2% positive. Five official splits, TEST once per split.

| Model | AUROC | AUPRC | Result |
|---|---|---|---|
| Best published (MTM, 2025) | 0.903 ± 0.020 | 0.583 ± 0.053 | State of the art on the official splits |
| Previous leaders (ViTST, GraFITi) | 0.892 / 0.891 | 0.531 / 0.566 | Published |
| Ours, 62,681 parameters | 0.916 ± 0.022 | 0.639 ± 0.039 | Win · AUPRC +0.056, beyond split spread |
| Ours without temporal memory | 0.900 ± 0.018 | 0.572 ± 0.037 | Below MTM on AUPRC |
| Trees on our statistics | 0.914 ± 0.020 | 0.618 ± 0.041 | Ours ahead on AUPRC, every split |

**The same temporal memory that won the event benchmarks wins clinical early warning.**

AUPRC is the clinically relevant metric at 4% prevalence. Temporal memory adds +0.072 AUPRC on every split. Reproduced on separate hardware.

# Beats the saved Transformer baselines at a fraction of the compute.

Wikipedia text (text8); same data and test set for all. Error: bits/char (bpc) or sMAPE %; lower is better.

| Comparison | Their error | Our error | Result and compute we used |
|---|---|---|---|
| vs Transformer, 4 layers, 4 passes (10M chars) | 1.908 bpc | 1.888 bpc | Better · 40% of its training, 18% of its inference compute |
| vs Transformer, 2 layers, 1 pass (10M chars) | 2.427 bpc | 1.955 bpc | Better · 96% of its training, 16% of its inference compute |
| vs LSTM-256 (10M chars) | 2.171 bpc | 1.955 bpc | Better at 88% of its inference compute (more training) |
| vs LSTM-512, 6 passes (10M chars) | 1.799 bpc | 1.888 bpc | Behind by 0.09; a wider model did not close it |
| vs best tuned LSTM (6 tuned runs) | 1.825 bpc | 1.888 bpc | Behind by 0.06; it used 92% of our compute |
| vs best tuned Transformer (4 tuned runs) | 1.996 bpc | 1.888 bpc | Better by 0.11 at ~equal compute (102%) |
| vs tuned Transformers (90M chars, ~2 PF) | 1.811 / 1.704 bpc | 1.783 bpc | Better than TF-256×4 at 1.06×; behind TF-192×4 |

**Wins vs both saved Transformers; a tuned small LSTM still leads. That gap is the next proof.**

Compute = estimated arithmetic operations. Baselines trained by us; 10 tuned runs at our two budgets complete, selected by validation. Scoreboard: evidence pack R1.

# Event-driven models aimed for parity. Ours beats small Transformers on compute.

Published event-driven and spiking language models, as reported by their authors, beside our completed result.

| Model | How it computes | Reported language result |
|---|---|---|
| EGRU on SpiNNaker2 (2023) | Clocked recurrent net; units fire events above a threshold | Matches LSTMs |
| SpikeGPT, 216M (2023) | Clocked binary spikes in a recurrent Transformer variant | “Competitive” with non-spiking models |
| Loihi 2 LLM, 370M (2025); sparse SSM (2026) | Conventional models made sparse/quantized; the chip supplies asynchrony | No accuracy loss / comparable to dense |
| Timing-coded LLM, 1.5B (2026) | Spike times carry values; closest to our races | “Clear gap” in language perplexity |
| Sleeping Machines (ours) | Learned arrival times choose which memories update; unchosen paths learn too | Beats two small Transformers at 40% of training compute; tuned LSTMs lead by 0.06 |

**To our knowledge, the first event-timed model to beat Transformer baselines on language with less compute, so far at small scale.**

Different datasets and scales; not a ranking. Ours: text8, 10M characters, small models, single runs, our baselines; LSTM-512 still ahead.

# Intelligence must compose across experience and execution.

A common substrate connects what a model learns with how and where it computes.

REPRESENTATION

## Meaningful inputs

Tokens, sensor events and typed comparisons enter through appropriate interfaces and act on shared persistent state.

LEARNING

## Reusable skills

End-to-end credit connects perception, memory, reasoning and action. Joint training tests transfer and retention.

EXECUTION

## Adaptive work

Learn timing and selective activity; recruit useful capacity for the task and the resources available.

**Small-data tables and large-data language are complementary tests of the same platform thesis.**

# One model family across today's architectural landscape.

Our approach combines strengths of today's main model types.

### Recurrent networks / Mamba

Exact member: learned delays over decaying memory (checked).

### Transformers

Exact member: races sample the softmax; delays give attention.

### Mixture of experts

Activate part of a large model.
We add private memory and timing.

## SLEEPING MACHINES / STATEFUL TEMPORAL PROGRAMS

One interface for content, timing, memory address and state, from single units to complete systems.
Target: computation through timing, sparse memory updates, small messages, and learning from unchosen paths.

### Statistics / retrieval

Counts and lookups of past data.
Combined with learned features.

### Adaptive memory

Models that adapt while in use.
With local, causal updates.

### Event-driven systems

Event-driven, neuromorphic systems.
We add trainable messages.

Mamba and attention containments are exact identities with numerical checks; other relationships are design relationships to be measured.

# Choose the right computation in each region.

One family can mix selective temporal programs with rich interaction and causal memory.

**01 / UNITS**

## Local programs

Each unit chooses what to remember, what message to send and when. Timing itself performs computation.

→

**02 / LAYERS**

## Adaptive work

A layer can activate one message or many: dense where needed, lean elsewhere.

→

**03 / COMPOSITIONS**

## Shared evidence

Learned features, counts and stored history can all inform the same prediction.

→

**04 / SYSTEMS**

## Same learner

The same learning rule works in batches or on a live stream, including learning after deployment.

**Counterfactual credit: learn from selected and unrealized routes**

**No global clock is required. Combined models, learning-after-deployment gains and chip energy are still milestones.**

# Messages race. Selected memories update. Alternatives learn.

A simple event loop connects representation, timing and sparse computation.

**01 / ENCODE**

## Signal + delay

A learned function produces a message and its arrival time from input and local memory.

→

**02 / SELECT**

## Temporal race

Arrival times select a receiver. Separate address keys and delivered values support sparse execution.

→

**03 / REMEMBER**

## Local state

Update the selected memory and propagate a new message into deeper computation.

→

**04 / LEARN**

## Alternatives

Use errors to improve message content and timing, including paths that were not selected.

**Counterfactual credit: learn from selected and unrealized routes**

**Implemented and trained today in ordinary software (CPU); no special hardware needed to run it.**

# Forward computation teaches more economical credit.

Real tokenized language: prospective credit accuracy and measured acquisition cost.

**FRESH-DATA CREDIT**

## 98.4% lower error

Eight new language prefixes; all eight improve over static credit. Fixed two-layer source and an adaptive lexical learner.

**MATCHED ACQUISITION**

## 2.09× faster

Versus exhaustive alternative learning. Context, route discovery, actual losses and factual correction included; 93.5% fewer supported operations.

**CONNECTED CONTROLLER**

## 77.8% less noise

Conditional gradient noise versus static credit in a connected controller with 37,377 parameters. Full factual correction preserves the exact mean gradient.

**Working online route-learning prototype. Next gate: better held-out prediction per total learning cost.**

# If this scales, adaptation becomes a routine capability.

Conditional economic and industrial opportunity; the proof path is explicit.

ADAPT AS DATA CHANGES

## Continual learning

Useful models could adapt more often as processes, language and environments change. Prove quality and complete cost on an online workload.

LEARN LOCALLY

## Machines + devices

Smaller learning-state budgets could support industrial sensing, logistics and embodied systems. Measure memory, latency and energy on target hardware.

BUILD A BROADER PLATFORM

## Software + hardware

Reusable learned credit could serve many routes and modalities, with a sparse asynchronous execution path. Verify deeper-model scaling and cross-domain transfer.

**Investment case: make useful adaptation economical, then turn demonstrated savings into a deployable runtime.**

# One family wins across different kinds of data.

One construction: memories that evolve with elapsed time, sparse addressed writes, typed comparisons, races. The P19 classifier reuses the event model’s temporal memory layer, same code and size.

| Data type | Benchmark | Result |
|---|---|---|
| Generative event streams | EasyTPP (5 public datasets) | Win on all 5: Taxi, Taobao, StackOverflow, Retweet, Amazon |
| One shared core, four event domains | EasyTPP, joint training (development) | One memory matches or beats separate models on 3 of 4 |
| Irregular clinical records | P19 sepsis, official splits | Win: AUPRC 0.639 vs 0.583 |
| Wearable sensors (activity) | PAM, official splits | Win: accuracy 0.978 vs 0.975 (4 of 5 splits ahead) |
| Temporal interaction graphs | TGB trade and wiki leaderboards | Win on both: 0.868 vs 0.863 (NAVIS); 0.835 vs 0.827 (TPNet) |
| Anonymous interleaved process logs | FAS v1, 3 seeds | Win vs six generic controls (0.592 vs 0.559) |
| Tokenized language (GPT-2 BPE) | FineWeb vs Kneser–Ney trigram | Ahead: 6.009 vs 6.537 (3 seeds); lead grows at 4× data |
| Character language, 10M | text8 vs tuned baselines | Win vs Transformers at ≤ equal compute; LSTM leads |
| Temporal reasoning, few examples | Structured event tasks | 99.7% vs Transformers 33–41% |
| Mixed-type tables | Synthetic; banknote | Learns the interaction; trees lead on real data |

**S2P2, the event-benchmark leader, is shown on one task type. We win on several.**

Large-scale language (90M characters) and real tables remain behind; the tokenized-language reference is a count model, a published small Transformer is next.

# Respect types. Race decisions. Compose learned state.

Meaningful comparisons become events before evidence enters a common neural representation.

RESPECT THE FIELD

## Typed comparisons

Compare a quantity with a threshold in its units. Test category membership, Boolean conditions and missingness explicitly. Category codes need no invented distance or order.

LEARN THE DECISION

## Temporal races

Comparison outcomes feed computational races and selected memory writes. Counterfactual consequences teach routes. Learned thresholds require branch, timing and factual-path credit; that extension is the next test.

COMPOSE THE EVIDENCE

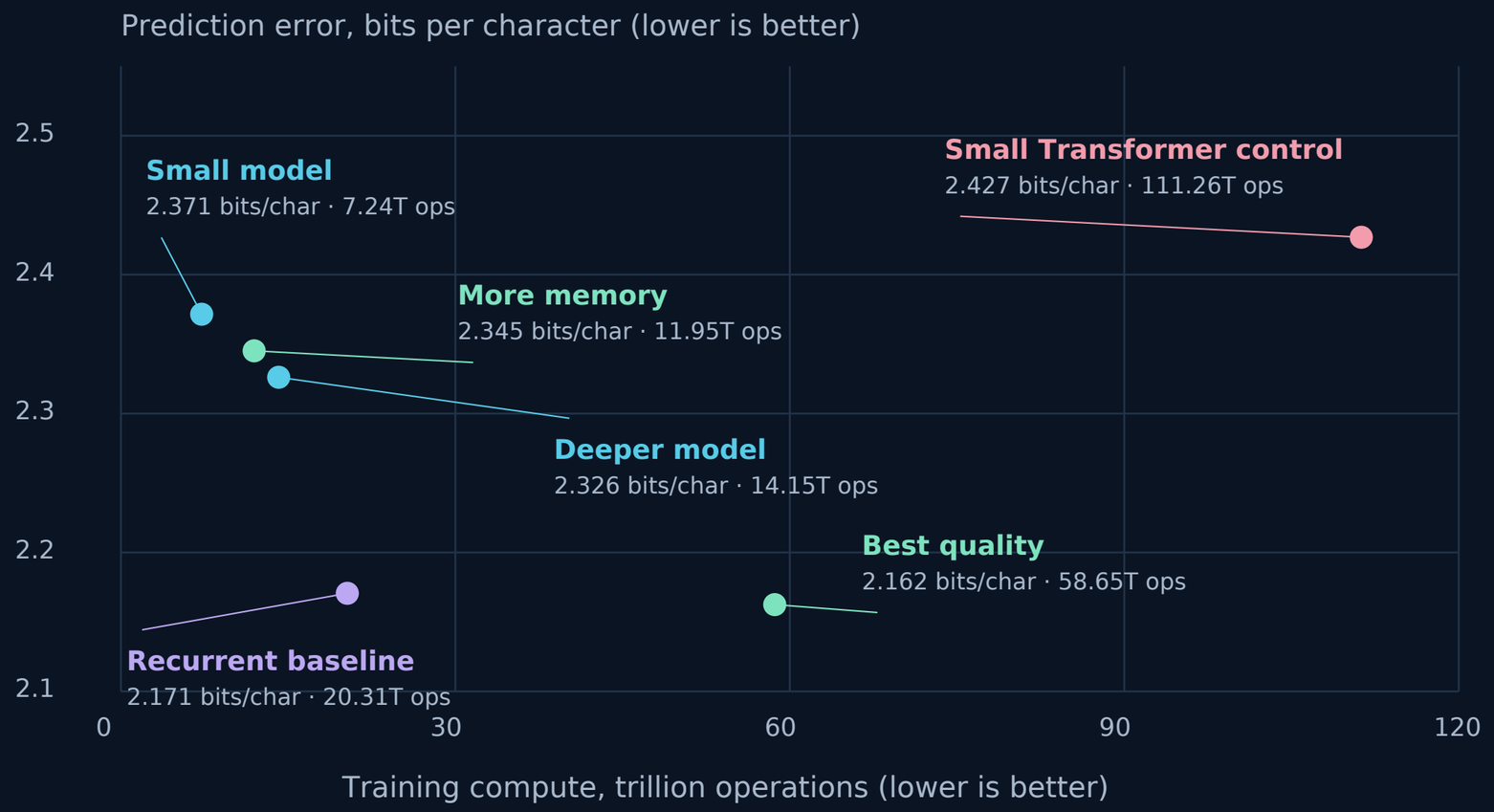
## Deep shared state

After a typed interface, learned vectors can mix meaningfully in persistent representations. The same event construction can connect tables, tokens, process traces and actions.

**Typed semantics + conditional selection + deep temporal learning. First mixed-type fit:100% synthetic DEV accuracy; fixed predicates.**

# Single pass: error versus training compute.

Each model sees the 10M training characters once. Lower and further left is better. Slide 2 adds multi-pass runs.




**BEST SINGLE PASS**

## 2.162 bpc

Edges LSTM-256 (2.171), with 2.9× its training compute.

**TRAINING COMPUTE**

## 15.4× less

Small model vs 2-layer Transformer: less compute and lower error.

Baselines trained by us on the same data and test set. Compute is estimated arithmetic (trillion operations), not measured energy or cost. Single runs.

# Start with low-batch language inference on existing hardware.

Product hypothesis for local and on-premise text applications; no customer interest reported yet.

WHO WOULD EVALUATE IT

## Platform teams

Teams deploying small language models under latency, memory or cost constraints. Demand validation is a next step.

WHAT WE WOULD SUPPLY

## Model + runtime

A trained model, sparse inference runtime and a reproducible quality-and-cost evaluation on the intended workload.

WHY THEY WOULD ADOPT

## Verified savings

Adoption depends on equivalent useful quality, service reliability and a measured reduction in complete deployment cost.

**Commercial gate: target ≥20% full-system cost reduction at matched quality on one selected workload.**

# Software proves the advantage. Hardware multiplies it.

One research program, with staged routes to deployment.

01 / MODELS + RUNTIME

## First product

Validate the language workload on existing hardware. License a model/runtime with auditable deployment economics.

02 / MEMORY-LOCAL ACCELERATORS

## 1.7–5.8× less data

Modeled per character: 1.7× fewer memory bytes than the best tuned Transformer and 5.8× fewer than the saved one, both beaten. 10× more memory adds 5% traffic. Tuned LSTMs: parity.

03 / CLOCKLESS SILICON

## Work only on events

No global clock: idle parts draw almost no power, answers arrive as soon as they are ready, and cool chips allow dense 3D stacking. Also the missing model for neuromorphic chips.

**The €3M round funds scale and a hardware cost model of the trained network; silicon follows measured proof.**

# A useful alternative could strengthen a hardware ecosystem.

Potential evaluation and licensing routes; no partner interest or relationship is claimed.

AMD

## Model + platform

Its ~$665M Silo AI acquisition and Liquid AI collaboration show the value of model/software capability on AMD hardware.

INTEL

## Event execution

Hala Point and Loihi 2 provide a relevant research environment for expressive event models and continuous learning.

GOOGLE

## Full-stack design

TPU model–hardware co-design creates a demanding evaluation venue for quality, memory and execution improvements.

**Partner proposition: demonstrate a differentiated workload advantage that improves platform adoption.**

# Price the runtime against verified customer value.

Illustrative annual license economics; assumptions to test with an evaluation partner.

| Per-customer assumption | Calculation | Annual amount |
|---|---|---|
| Eligible workload cost | Existing deployment cost we can actually influence | €1.00M |
| Gross savings | 20% complete-cost reduction at matched quality | €200k |
| Runtime license | 20% of verified gross savings | €40k |

Customer retains €160k/year before migration and other costs. At the same assumed price, 25 deployments would generate €1M annual license revenue.

**Validate pricing through matched quality, reliable operation and measured customer benefit.**

# AI research and software architecture in one founder.

Tero Keski-Valkama · Sole founder · Public career and invention record.

## Tero Keski-Valkama

### Sole founder

Experience spans applied ML, multimodal foundation models and software architecture. The round adds specialist capacity around the founder.

| | |
|---|---|
| **2012–2018** | **Cybercom · Software Architect**<br>Historical role in public professional profile. |
| **2018–2022** | **HERE · Lead ML & AI Engineer**<br>Self-reported research and applied ML leadership. |
| **2022–2025** | **Alloy.ai → Kaiko.ai**<br>Staff engineering; multimodal foundation-model work. |
| **2019–2025** | **35 patent publications · Inventor**<br>US and EP patents; assignee HERE Global. |

Historical roles are self-reported; inventorship is documented. Planned hiring: learning research, runtime engineering, evaluation and measurement.

# Build defensibility across the model and its execution.

The strongest potential asset is an integrated stack that is difficult to reproduce and deploy.

MODEL + LEARNING

## Research know-how

Timing, state and alternative-credit methods, supported by retained ablations and numerical contracts.

RUNTIME + HARDWARE

## Execution know-how

Sparse kernels, memory management, measurement and hardware mapping can reinforce the model advantage.

RIGHTS + PATENTS

## Company assets

Secure company ownership. Assess technical inventions for selective patent filings; protect confidential implementation know-how.

**Funding builds company-owned IP: ownership review, selective patent filings and confidential know-how.**

# Three gates determine whether this becomes a business.

Each gate has a falsifiable result and a consequence for further spending.

01 / QUALITY AT SCALE

## Replicate

At least three runs per result, fixed held-out tests and tuned baselines. Compare quality and total compute; publish failures too.

02 / COMPLETE ECONOMICS

## Measure

Show the fast sparse runtime gives the trained model’s answers. Measure serving cost, latency, memory and energy on one workload.

03 / ADOPTION

## Evaluate

Identify a use case and an evaluator willing to test it. Confirm useful quality, integration effort and willingness to pay.

**If a gate fails, diagnose or narrow the workload before committing more compute or new silicon.**

# The next 18 months produce evidence investors can underwrite.

Proposed milestones measured from funding; spending advances with completed gates.

**MONTHS 0–3**

## Team + proofs

Hire the initial team. Resolve IP ownership and review priority filings. Verify the trained sparse runtime; repeat larger-data results.

**MONTHS 3–6**

## One workload

Run tuned controls and measure total cost, latency, memory and energy. Select a workload only if quality and economics hold.

**MONTHS 6–12**

## External trial

Seek independent reproduction and prospective customer evaluations. Test pricing, integration effort and willingness to pay.

**MONTHS 12–18**

## Product + scale

Turn a validated workload into a deployable runtime. Expand models and assess hardware mapping for the next financing stage.

**Next-round objective: replicated quality, a measured cost advantage and a credible evaluation pathway.**

USE OF FUNDS

# €3M funds an 18-month research and engineering program.

Team, staged compute, hardware feasibility and IP protection, including selective patent filings.



| | |
|---|---|
| Team: research + systems | **€1.35M \| 45.0%** |
| Compute + independent replication | **€0.90M \| 30.0%** |
| Hardware feasibility + measurement | **€0.25M \| 8.3%** |
| IP, legal, operations | **€0.20M \| 6.7%** |
| Contingency reserve | **€0.30M \| 10.0%** |

**€2.70M operating plan + €0.30M reserve · Planned operating burn averages €150k/month.**

Budget assumptions: six average FTE including founder at €150k/year fully loaded; compute and hiring quotes remain to be obtained.

# Raise €3M to establish a commercially valuable platform.

Proposed priced round; terms remain subject to company formation and investor diligence.

**CAPITAL TO RAISE**

## €3M

18 months of research, systems engineering and staged validation.

**PROPOSED PRE-MONEY**

## €50M

A premium research-stage proposal for the model/runtime/substrate opportunity.

**INITIAL OWNERSHIP**

## 5.66%

€53M post-money, before option-pool changes, preferences and later dilution.

The proposal prices a model/learning family, sparse runtime and future chip architecture that already beats Transformer baselines at a fraction of the compute. A confirmed 90M-character win and a NeuroBench first place would justify a higher next price (the €100M scenario in the appendix).

**Valuation sensitivities and conditional exit scenarios are in the appendix.**

# Build the model. Prove the economics. Open the hardware path.

€3M proposed raise · €50M proposed pre-money · Private investor discussion.

# €3 million

Already beating Transformer baselines on compute.
This round tests whether the advantage scales.

**01** Replicated language quality and learning mechanisms.

**02** A deployable runtime with measured system economics.

**03** External evaluation and a credible path to hardware.

Tero Keski-Valkama · Private investor diligence available by arrangement

# The same model improves when unchosen paths receive credit.

Controlled small-model comparison; held-out error and estimated complete training work.

**TEST BITS / CHARACTER**

## 2.506 → 2.371

Timing-only credit versus added local alternative-value route credit.

**ESTIMATED FITTING WORK**

## +0.30%

Hard forward selection and parameter count retained.

**CHARACTER PERPLEXITY**

## −8.9%

Derived from the BPC difference; one controlled seed.

## Same small model; hard selection retained

A deeper variant also improves by 0.130 bits per character. Both are single-seed comparisons; numerical details and model definitions follow.

**A falsifiable architectural mechanism, with a small counted arithmetic increment.**

# More available state. The same eight selected writes.

Small model with alternative credit: available memory doubles; selected updates stay fixed.

## Pool 2



16 slots · 512 value scalars
8 writes · 16 keys scored
2.371 bpc

## Pool 4



32 slots · 1,024 value scalars
8 writes · 32 keys scored
2.345 bpc

**WHAT SCALES**

## 8 writes

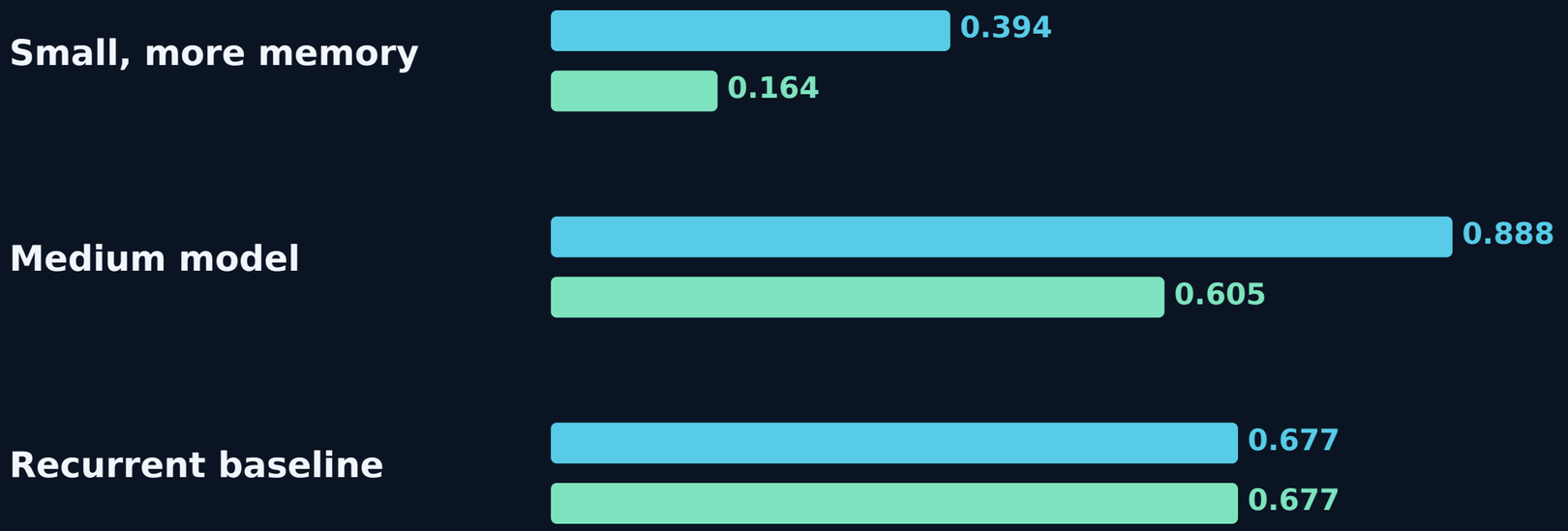
Winner-only arithmetic +0.66%. Fitting work +65.0%. Keys and value storage double.

**Quality improves by 0.026 bpc; total capacity and learning costs remain explicitly charged.**

# The sparse path is implemented; system proof is next.

Inference arithmetic per predicted character. Green bars: our implemented sparse runtime, which computes only the selected paths.

Inference compute per character, million operations (lower is better)




**STATUS**

Every memory address is still scored. Checking that the sparse runtime reproduces the trained model exactly is pending, as are measured traffic, latency and energy.

Cyan: full computation (all paths) Green: sparse runtime, only the selected paths (the baseline has no sparse mode)

Arithmetic estimates, not measured service cost. Our evaluation also pays about two extra positions of warm-up per predicted character; the LSTM evaluates differently.

# An adaptation signal exists. Its work is accounted for.

Predict-before-update experiment on a new text stream; separate from the latest model family.

**PREDICT BEFORE UPDATE**

## 3.191 → 3.096

8,191 new development targets; same persistent state, frozen versus full-backbone adaptation.

**COSTED LEARNING**

## 10.7× work

Total online processing work includes credit, backward and optimizer. Retention under drift and native asynchronous learning remain open.

**Online-learning upside: persistent adaptive systems. Current result: one costed pilot.**

# One substrate can support new communication and learning patterns.

Application hypotheses with first proofs; no codec, embodied-transfer or AGI result yet. The first row needs no special hardware.

| Opportunity | Architectural fit | First discriminating proof |
|---|---|---|
| Event-log anomaly detection (existing hardware) | Built-in biases: computational time, first-arrival races, separate state per concurrent process. | FAS v1 is solved by a simple route rule (AUROC 0.755 at 256 events vs our 0.600); proof needs a harder v2 with ambiguous interleaving. |
| Irregular inputs | Addressed events retain time and update persistent programs. Dense regions remain available. | Same causal data/queries; compare timestamp-aware controls and burst/idle resource use. |
| Asynchronous codecs | Learn what, where and when to send; decoder retains predictive state. | Actual bytes versus distortion/task quality, sender/receiver work and causal latency. |
| Memory meta-learning | Slow rules learn fast writes, retention, retrieval and scheduling from later utility. | Held-out delayed-use tasks; complete useful-return credit against immediate utility. |
| Embodied cognition | Shared sensing/action/language paths can exchange learned physical abstractions. | Motor-to-cognitive and reverse transfer measured separately; matched data/work and isolated-path controls. |

# The contribution is the complete construction.

Established alternatives already address parts of this problem; they remain essential controls.

| Approach | Established strength | Our comparison obligation |
|---|---|---|
| Transformer / MoE | Strong quality; MoE separates available parameters from selected expert work. | Match quality/data and charge routing, memory traffic and training. |
| RNN / selective SSM | Persistent state and efficient sequence processing. | Test whether temporal races add useful quality or allocation. |
| Neuromorphic systems | Event-driven hardware and local learning precedents. | Prove expressive model quality, mapping and measured energy. |
| Sleeping Machines | Joint timing, sparse memory, deep representation and alternative credit. | Scale integrated mechanisms without replacing them with dense carriers. |

**Sparsity, recurrence and on-chip learning alone are not the novelty claim.**

# What clockless, memory-local silicon can buy.

Physical mechanisms, their basis and what must be measured. No chip exists yet.

| Benefit | Mechanism | Proof still needed |
|---|---|---|
| Energy | Cost model: 1.7–5.8× fewer bytes/char than the Transformers we beat; tuned LSTMs at parity; no clock tree. | Quality at large pools; chip-level measurement. |
| Latency | Average-case completion; events propagate when ready; races resolve at first arrival. | Event-trace latency on FPGA semantics prototype. |
| Cooling & density | Low average power relieves the power-density limit: memory-on-logic 3D stacking, chiplets without a global timing domain. | Activity and peak-burst profiles on real streams. |
| Communication | Small messages sent only when events occur, not full vectors every clock: wire energy and wiring area track event rate. | Measured message rates and header overhead. |
| Local memory | Addressed persistent state maps to banks beside compute; non-volatile memory makes dormant capacity near-free. | Locality and reuse-distance profile of trained models. |
| On-chip learning | Route credit and delay/gate updates are local quantities. | Event-local learning rule matching current trainer quality. |
| Also | Near-threshold voltage, native event sensors, low EMI, defect tolerance. | Hardware partner evaluation. |

**The neuromorphic hardware class exists; it lacks a competitive trainable model. That is the opening.**

# AMD’s Silo AI deal supports the strategic logic.

Announced approximately $665M all-cash acquisition in 2024; a mature-company precedent.

**WHAT AMD ACQUIRED**

## Team + stack

AI scientists and engineers, enterprise solutions, model development and software capability on AMD hardware.

**THE LUMI CONNECTION**

## Poro / Viking

AMD highlighted multilingual Poro and Viking models trained on AMD-powered LUMI, with university and supercomputer collaborators.

**OUR LESSON**

## Prove leverage

Build a capability that improves a platform’s economics and adoption. One small research project is not a direct pricing comparable.

**Efficient models can matter to a chip ecosystem; training on its chips is one part of the value case.**

# The constraints reach from power grids to robot batteries.

Economic context for efficient intelligence; these figures are not our revenue market.

**DATACENTERS**

## 485 → 950

TWh/year, 2025 → 2030 central IEA projection. Power and data movement make system efficiency economically important.

**MOBILE**

## 5.8B

Unique mobile subscribers in GSMA’s 2026 report. On-device inference creates a prospective route to broad distribution.

**ROBOTICS**

## 5M

Industrial robots operating in 2025. Sparse sensing, persistent state and adaptation are prospective use cases.

**Select one measurable initial workload; expand only after quality and system gains hold.**

# Successful platform adoption can support very large outcomes.

Conditional commercial arithmetic illustrates the scale required; no forecast or market multiple claim.

| Successful outcome assumption | Value capture assumption | Equity-value scenario |
|---|---|---|
| €10B/year eligible compute spend | 20% savings × 20% capture = €400M/year; 25× revenue assumption | €10B |
| €50B/year eligible compute spend | 20% savings × 25% capture = €2.5B/year; 20× revenue assumption | €50B |

A cross-platform substrate could participate in cloud inference, persistent agents and embodied systems. Reaching these outcomes requires sustained quality, scale, margins and broad adoption.

**The upside is large if the platform works and wins adoption. Its probability remains unmeasured.**

# The completed comparisons, using consistent resource units.

Frozen 10M-character, one-pass evidence. Model definitions are on the next slide.

| Model | Error, bpc ↓ | Parameters (K) | Training, TFLOPs ↓ | Train MFLOPs/char ↓ | Full run MFLOPs/char ↓ | Sparse run MFLOPs/char ↓ |
|---|---|---|---|---|---|---|
| Small: timing-only credit | 2.506 | 108.9 | 7.22 | 0.722 | 0.240 | 0.163 |
| Small: alternative credit | 2.371 | 108.9 | 7.24 | 0.725 | 0.240 | 0.163 |
| Small: more memory | 2.345 | 177.0 | 11.95 | 1.196 | 0.394 | 0.164 |
| Deeper: alternative credit | 2.326 | 210.0 | 14.15 | 1.416 | 0.463 | 0.310 |
| Medium: alternative credit | 2.183 | 422.5 | 26.79 | 2.680 | 0.888 | 0.605 |
| Medium: more memory | 2.179 | 689.8 | 44.07 | 4.410 | 1.458 | 0.607 |
| Best quality: wider model | 2.162 | 940.9 | 58.65 | 5.868 | 1.947 | 1.324 |
| Recurrent baseline (LSTM) | 2.171 | 338.4 | 20.31 | 2.032 | 0.677 | 0.677 |
| Small Transformer control | 2.427 | 1658.9 | 111.26 | 11.133 | 3.710 | 3.710 |

↓ = lower is better. bpc = bits per character of prediction error. TFLOPs/MFLOPs = trillion/million arithmetic operations (estimates). Train = per training character; run = per predicted character. Full run computes every path; sparse run only the selected ones (exact-match check on trained models pending). Full 15-row CSV retains older models.

# Plain-language labels map to reproducible model records.

Message width counts scalars; available memory counts local state slots.

| Deck label | Width / layers | Memory slots | Record configuration |
|---|---|---|---|
| Small model | 32 / 4 | 16 | p32 / d4 / pool2 |
| Small: more memory | 32 / 4 | 32 | p32 / d4 / pool4 |
| Deeper model | 32 / 8 | 32 | p32 / d8 / pool2 / skip2 |
| Medium model | 64 / 4 | 16 | p64 / d4 / pool2 |
| Medium: more memory | 64 / 4 | 32 | p64 / d4 / pool4 |
| Best quality | 96 / 4 | 16 | p96 / d4 / pool2 |
| Saved controls | 256 hidden width | Conventional architectures | LSTM-256; Transformer-256 × 2 layers |

# The comparison boundaries are part of the evidence.

Maintain the original records and disclose what a result actually measures.

| Boundary | Current completed evidence | Remaining work |
| --- | --- | --- |
| Data / evaluation | Frozen table: 10M training characters. Report: completed 90M run, 1.8573 bpc on test characters 95M–96M, 256-character windows. | Align modern controls, target coverage and fit work; full official 5M test remains open. |
| Learning / seeds | 10M ledger: 1,220 updates, lr .004 cosine; one seed; state reset per fit segment. | Replication, robust recipe, depth/horizon and matched larger-data comparisons. |
| Sparse execution | All keys scored; selected values only in winner evaluator; small random FP64 equality fixtures. | Actual trained FP32 winner/state/cache equality and held-out rescore. |
| System economics | Arithmetic traces/formulas; persistent local state within windows. | Setup, warmup, traffic, memory, throughput, latency, joules and total cost. |
| Adaptive workload | Fixed scored queries; declare late/missing readout, feedback and fallback. | Charge attempted work; report coverage separately if abstention is the task. |

# Keep the failures beside the positive mechanism results.

A disciplined research program should revise interpretations when evidence changes.

| Finding | Interpretation | Response |
| --- | --- | --- |
| Uncredited D8 pool expansion worsens quality | More capacity needs learnable selection; it is not automatically useful. | Retain negative pair; test credited expansion with matched depth/seed. |
| Write-credit variants regress or diverge | A particular local estimator is unstable; not all route credit is equivalent. | Withdraw variants; preserve original result and derive corrected credit. |
| Best p96 costs more than LSTM | Small quality lead is not comparable-quality resource superiority. | Tune and scale; require complete quality–cost frontier. |
| Event-vision screening is mixed | Small coarsening gains do not clear every full-screen seed or strong RBF controls. | Avoid cross-domain supremacy claim; keep modality-specific gates. |

# Reverse-underwriting shows the assumptions behind a price.

This arithmetic is a sensitivity tool; the research does not estimate platform-success probabilities.

**BULLISH PRICED ROUND**

## €50M

€3M raise → €53M post-money; 5.66% new-investor ownership.

**REVERSE UNDERWRITING**

## 6.74%

Required platform-success probability in the stated €10B exit scenario.

**STRETCH SCENARIO**

## €100M

€100M pre-money requires 13.49% in the same model; 2.91% ownership.

**V = p × exit equity × future retention / (1 + discount rate)^years**

Assumptions: €10B exit; 30% stake retention; 10 years; 15% discount; failure value zero.

**Required odds are reverse-calculated conditions. The benchmarks do not estimate those probabilities.**

# The price depends on dilution, exit scale and time.

Required platform-success odds at 10 years and 15% discount; zero failure value.

| Exit equity value | Investor stake kept after dilution | Success odds needed at €50M | Success odds needed at €100M |
|---|---|---|---|
| €3B | 10% | 67.43% | 134.9% (infeasible) |
| €3B | 30% | 22.48% | 44.95% |
| €3B | 50% | 13.49% | 26.97% |
| €10B | 10% | 20.23% | 40.46% |
| €10B | 30% | 6.74% | 13.49% |
| €10B | 50% | 4.05% | 8.09% |
| €30B | 10% | 6.74% | 13.49% |
| €30B | 30% | 2.25% | 4.50% |
| €30B | 50% | 1.35% | 2.70% |

Sensitivity assumptions only. Probabilities above 100% are infeasible in this model. Preferences, option pools and intermediate financing cash flows are omitted.

# A private evidence pack and primary external context.

Source labels in slide footers are clickable; companion files must travel with the PDF.

**R1** **Private completed language evidence ledger (3 October 2026)**

Saved native fits and E64 controls. Original JSON records and SHA256 hashes are supplied in the private diligence pack.

**R2** **Private online-learning experiment, 8,191 targets**

Predict-before-update pilot: quality and complete processing-operation estimates; one checkpoint, stream and learning rate.

**R3** **Architecture, numerical contracts and report**

Private technical report: whole-family definition, landscape, composition and scoped evidence. Modern matched controls, trained sparse parity and measured hardware energy remain open.

**M1** **IEA — Key Questions on Energy and AI, executive summary**

485 TWh datacenter electricity in 2025; central projection 950 TWh in 2030. Sector context, not addressable revenue.

**M2** **GSMA — The Mobile Economy 2026**

5.8 billion unique mobile subscribers and 8.8 billion connections. Subscribers are not annual device sales or customers.

**M3** **IFR — World Robotics 2026 release, 24 September 2026**

5 million operating industrial robots in 2025 and more than 600,000 annual installations; no adoption by this project.

**T2** **Fedus et al. — Switch Transformers**

Capacity beyond selected computation has established MoE precedent; sparsity alone is not a novelty claim.

# Public precedents establish relevance, not endorsement.

Historical roles are self-reported; inventorship is checked against the published patent record.

**S1** **AMD — Silo AI acquisition completion, 12 August 2024**

Announced approximately $665M all-cash acquisition; Poro and Viking on AMD. Mature-team strategic precedent, not a pre-seed valuation comparable.

**S2** **AMD — Acquisition announcement and LUMI collaboration, 10 July 2024**

AMD highlighted enterprise solutions, expertise, software and Poro/Viking training on LUMI using AMD hardware; no single exclusive acquisition cause established.

**S3** **Liquid AI — $250M Series A announcement, 13 December 2024**

Financing announcement and AMD collaboration on efficient models; funding amount is not a disclosed company valuation.

**S4** **AMD Ventures — investment focus**

AI models, infrastructure and silicon investment focus; published Series A–C focus means this project may be earlier than typical admission.

**S5** **Intel — Hala Point research system, 17 April 2024**

Loihi 2 neuromorphic research prototype, event computing and prospective continuous learning. No Intel interest in this project claimed.

**S6** **Google — Eighth-generation TPU architecture announcement**

TPU 8t/8i, model–hardware co-design and sparse MoE infrastructure. Existing internal capability creates both potential fit and competition.

**F2** **Patent record — Google Patents inventor search (all publications)**

35 published US and EP patent documents name Tero Juhani Keski-Valkama as inventor or co-inventor (priority 2019 onward; publications through October 2025); assignee HERE Global B.V. Experience evidence, not Sleeping Machines-owned IP.

# Prior event-driven language models.

Summaries of each paper's own claims; datasets, scales and compute conventions differ from ours.

**P1** **Nazeer et al., Language Modeling on a SpiNNaker 2 Neuromorphic Chip (2023)**

Event-based GRU on a neuromorphic chip; authors: first neuromorphic language model to match LSTMs.

**P2** **Zhu et al., SpikeGPT: Generative Pre-trained Language Model with Spiking Neural Networks (TMLR 2024)**

216M-parameter spiking language model; authors: competitive with non-spiking models; fewer operations projected on neuromorphic hardware.

**P3** **Abreu et al., Neuromorphic Principles for Efficient Large Language Models on Intel Loihi 2 (2025)**

370M MatMul-free model quantized for Loihi 2; no accuracy loss versus its unquantized version.

**P4** **Richter et al., Event-Driven Language Models with Sparse Neural Activity for Neuromorphic Hardware (2026)**

Thresholded activations in quantized linear-attention models; comparable to dense models with up to 4x fewer effective operations.

**P5** **Zhao et al., Large Language Models with At Most One Spike per Neuron (2026)**

Time-to-first-spike coding on GPT-2/BERT, up to 1.5B parameters; authors report a clear perplexity gap on language modeling.

# The language experiments establish three useful facts.

Wikipedia text (text8), 10M characters of training. Prediction error in bits per character: lower is better.

### IT LEARNS REAL TEXT

## 1.888 bits/char

Better than the 4-layer Transformer baseline (1.908) with 40% of its training compute. The larger LSTM (1.799) is still ahead.

### TRAINING UNCHOSEN PATHS HELPS

## −0.135 bits/char

Giving learning signal to the paths the model did not take cuts the error of the same small model, for 0.3% more training compute.

### MORE MEMORY, SAME RUNNING WORK

## 16 → 32 slots

Doubling memory slots lowers the error while each input still updates only 8 slots. Training cost rises; running cost barely changes.

**Single runs on one dataset. The next proof is repeated runs plus a measured cost advantage.**